

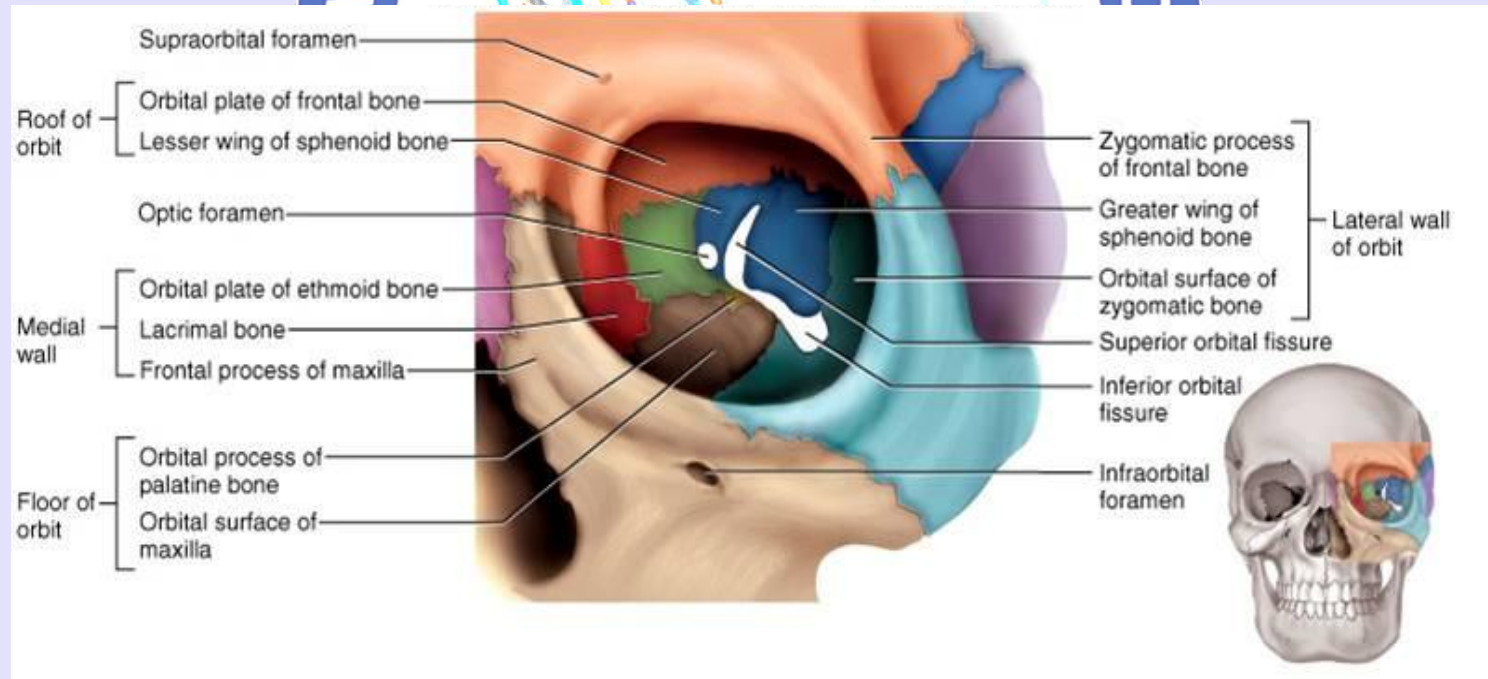
Orbital Cavity.

(boundaries and extraocular muscles)



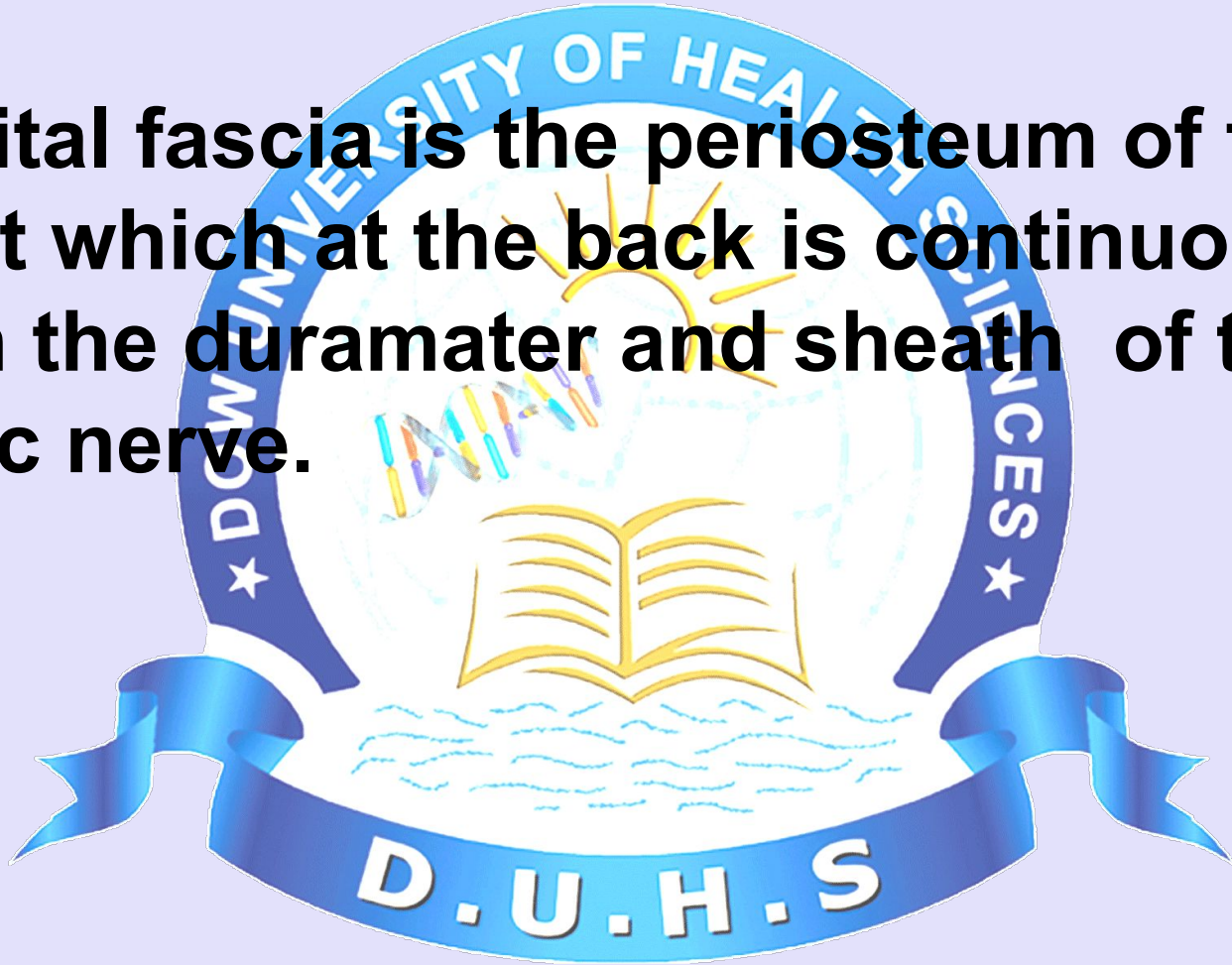
THE ORBIT

- Orbit is a bony cavity shaped like a four-sided pyramid
- Lying on each side of the root of the nose
- Lodges the eyeball.



Orbital fascia

- **Orbital fascia is the periosteum of the orbit which at the back is continuous with the duramater and sheath of the optic nerve.**



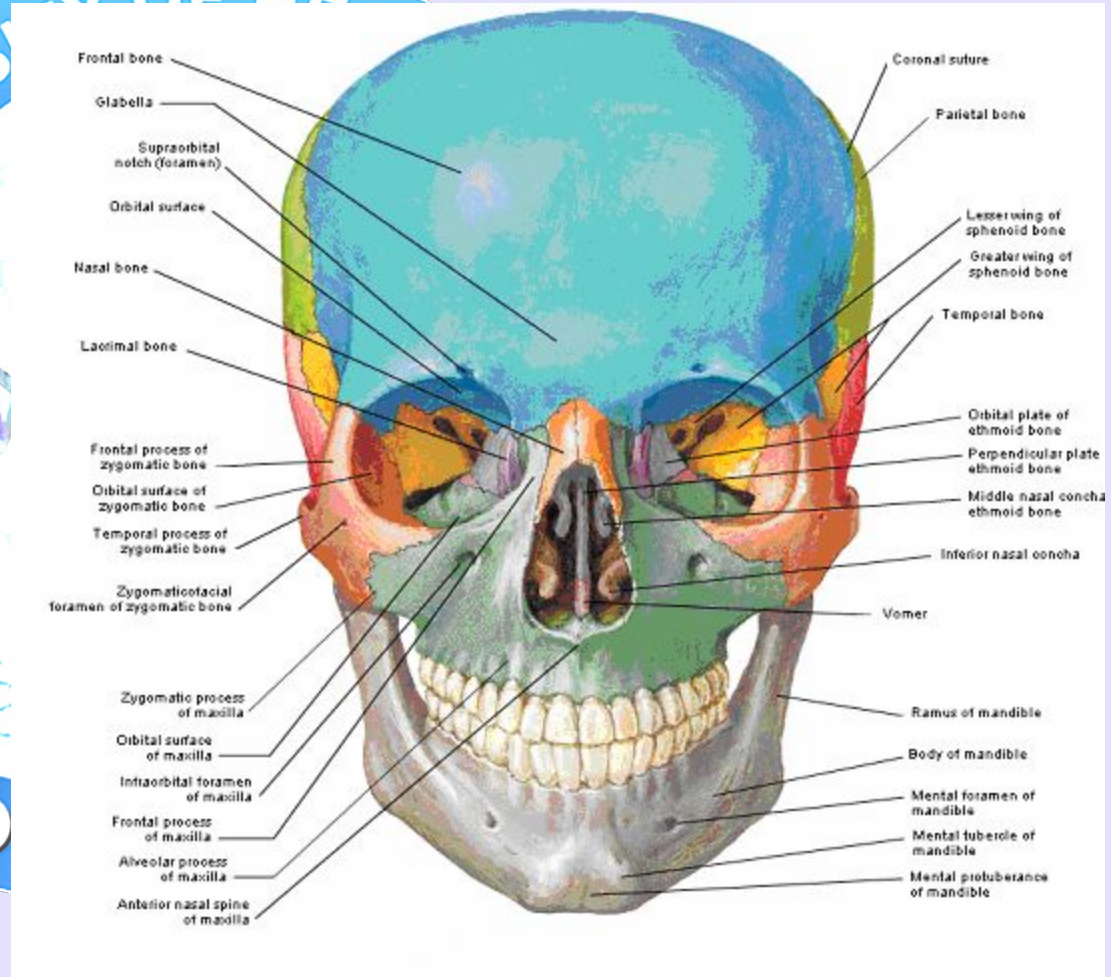
RELATIONS

SUPERIORLY

- Ant. Cranial fossa
- Meninges
- Frontal lobe of the cerebral hemisphere

MEDIALY:

- Nasal cavity
- Ethmoidal sinus
- Sphenoidal sinus



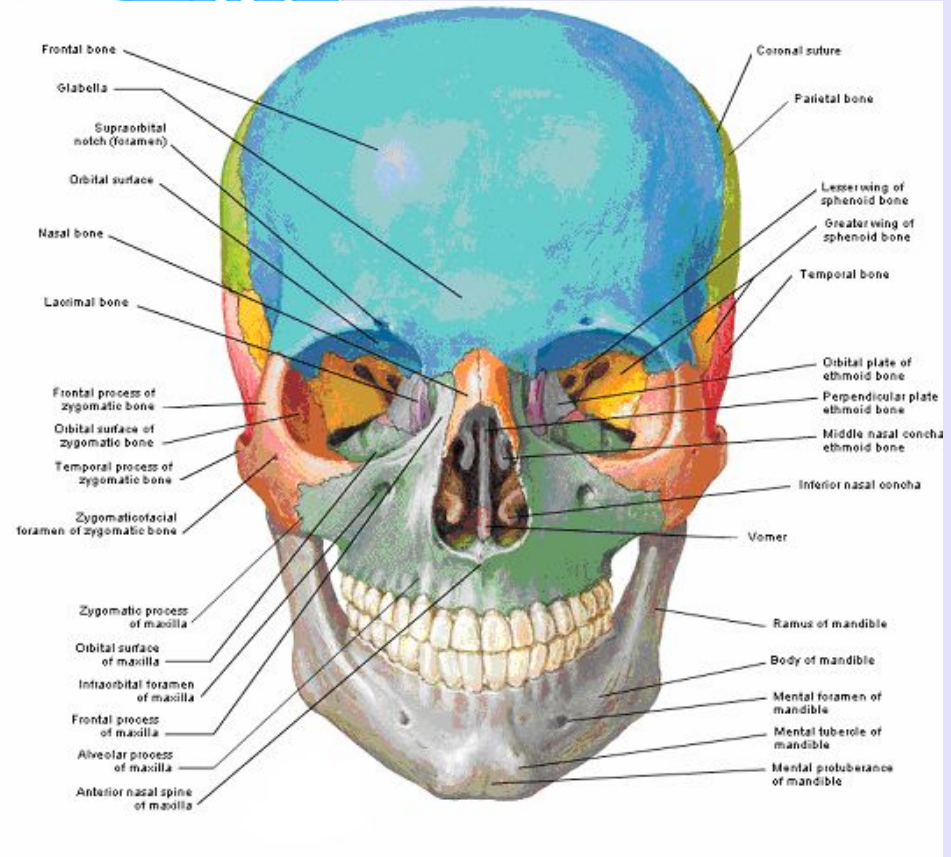
Contd...

INFERIORLY:

- Maxillary sinus.

POSTERIOROLATERALLY:

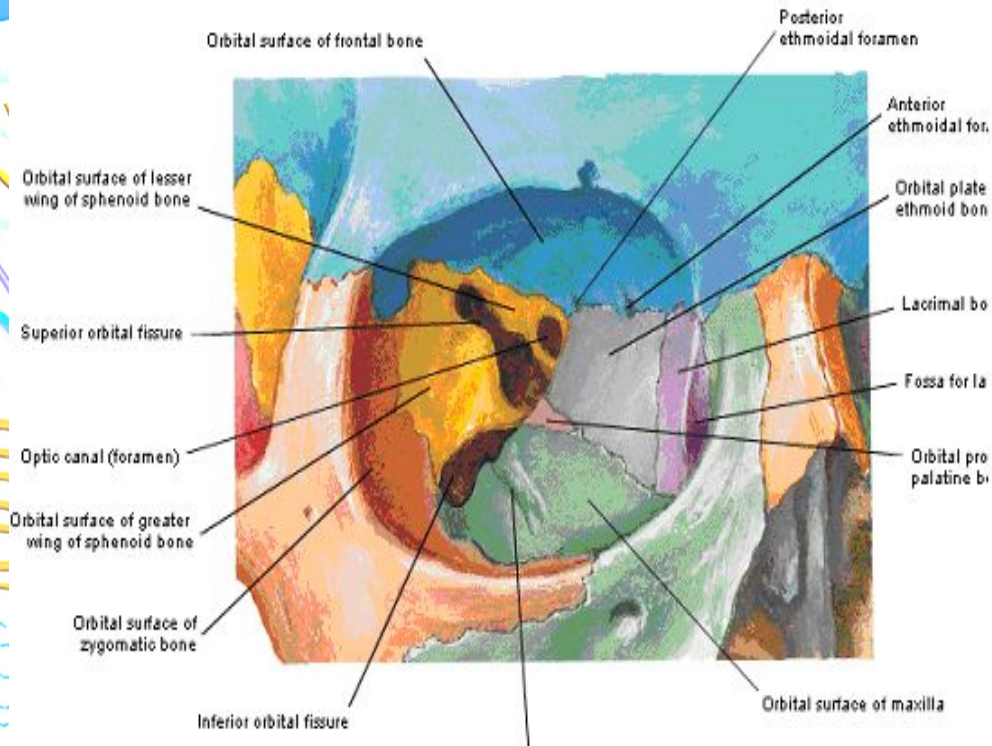
- Infratemporal fossa.
- Middle temporal fossa.



Boundaries

The orbit has

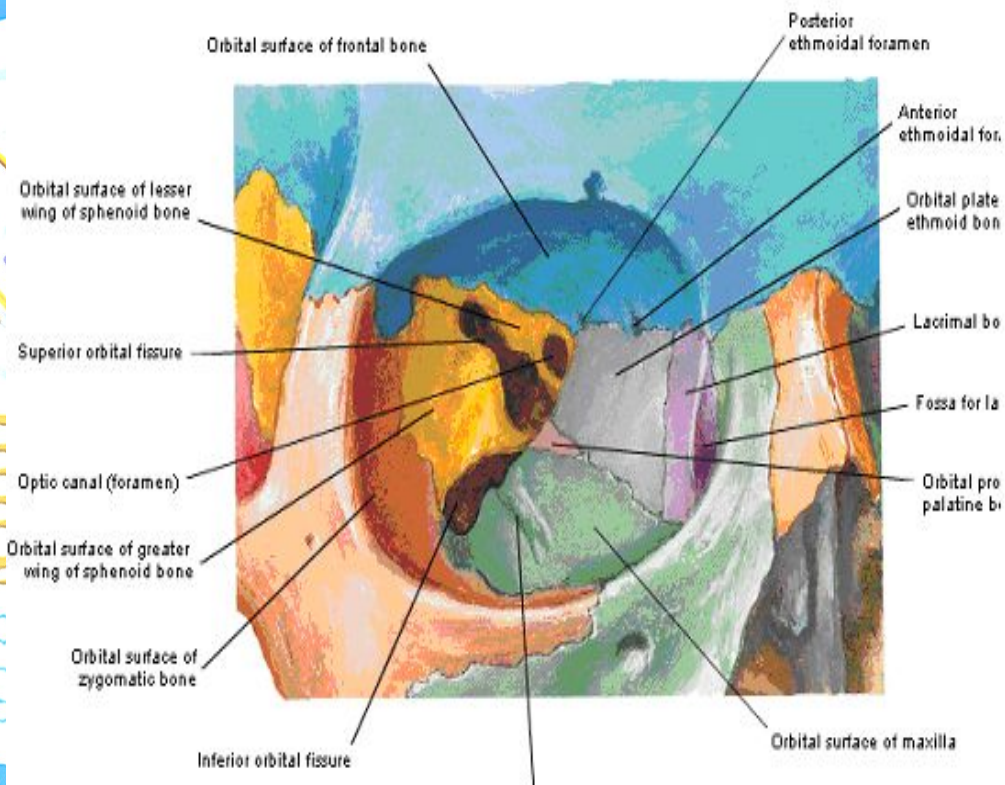
- **An apex**at the posterior end
- **A base**.....orbital margins
- **4Walls**....Roof, Floor, Lateral & Medial walls



ROOF

Formed by

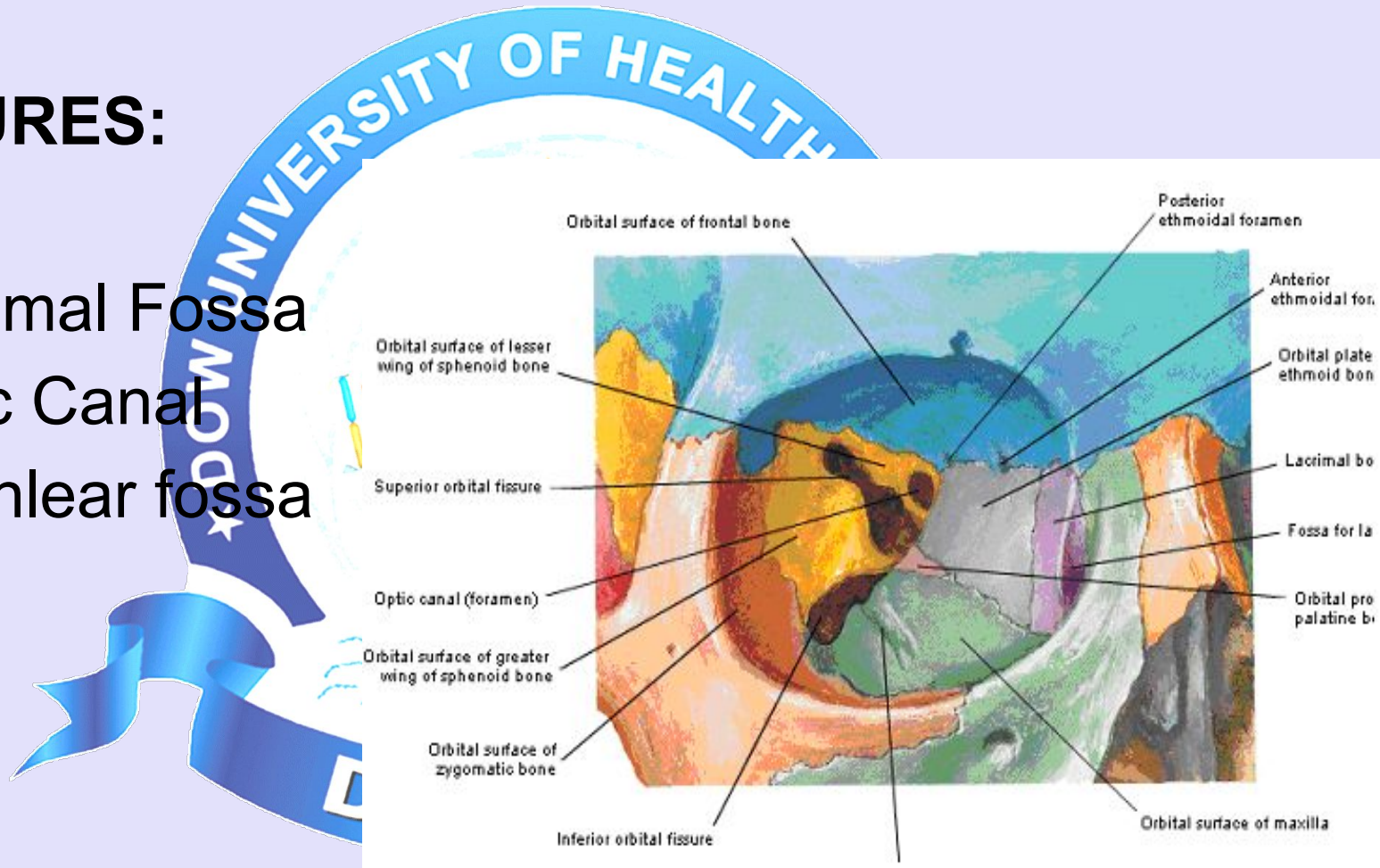
- Orbital part of the frontal bone
- Lesser wing of the sphenoid bone
- The roof separates the orbit from the anterior cranial fossa.



Contd...

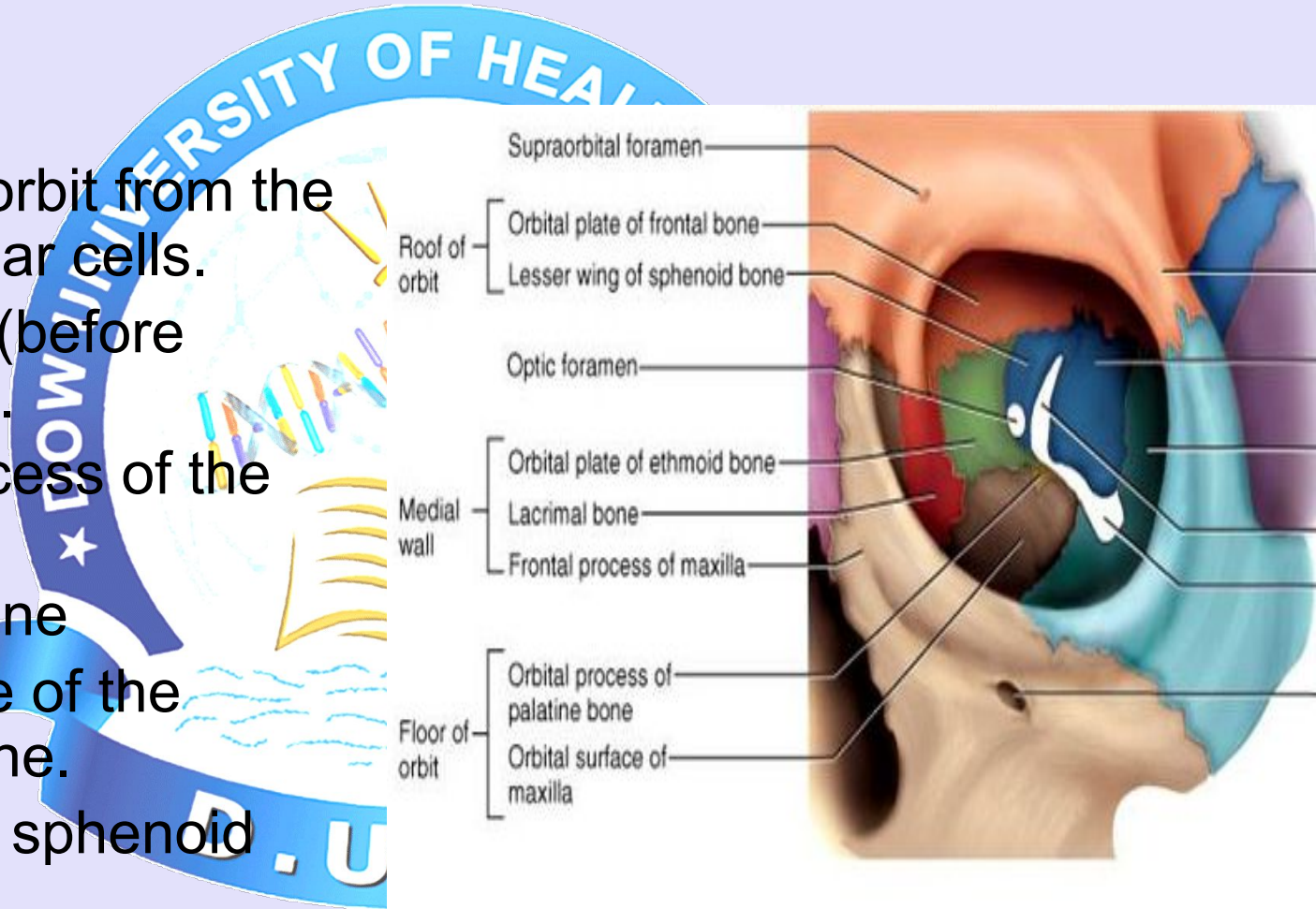
FEATURES:

- Lacrimal Fossa
- Optic Canal
- Trochlear fossa



MEDIAL WALL

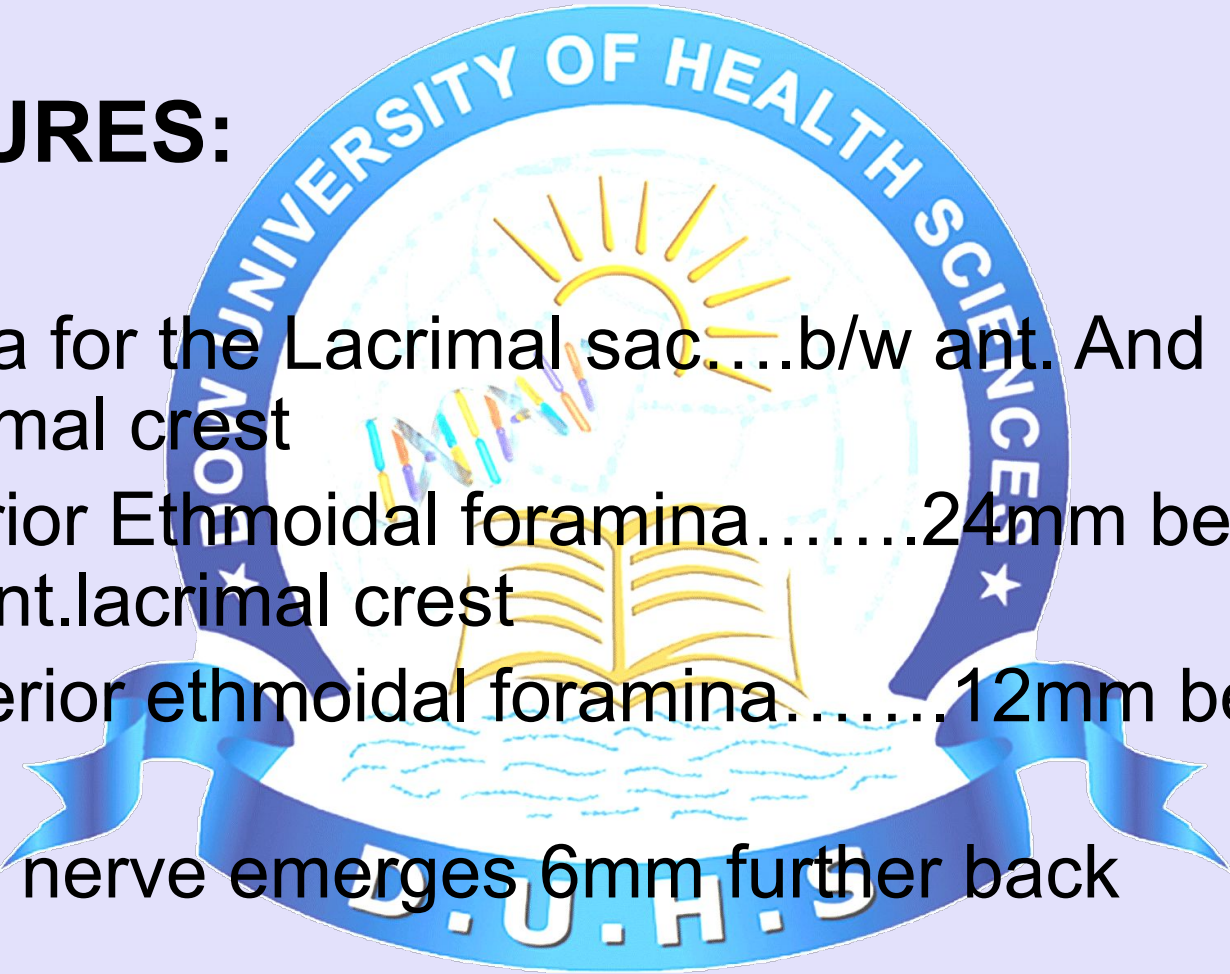
- 5 cm long.
- Separates orbit from the ethmoidal air cells.
- Formed by (before backwards).
- Frontal process of the maxilla.
- Lacrimal bone
- Orbital plate of the ethmoid bone.
- Body of the sphenoid bone.



Contd....

FEATURES:

- Fossa for the Lacrimal sac....b/w ant. And post. Lacrimal crest
- Anterior Ethmoidal foramina.....24mm behind the ant.lacrimal crest
- Posterior ethmoidal foramina.....12mm behind this
- Optic nerve emerges 6mm further back



LATERAL WALL

- Thickest wall
- 5cm long

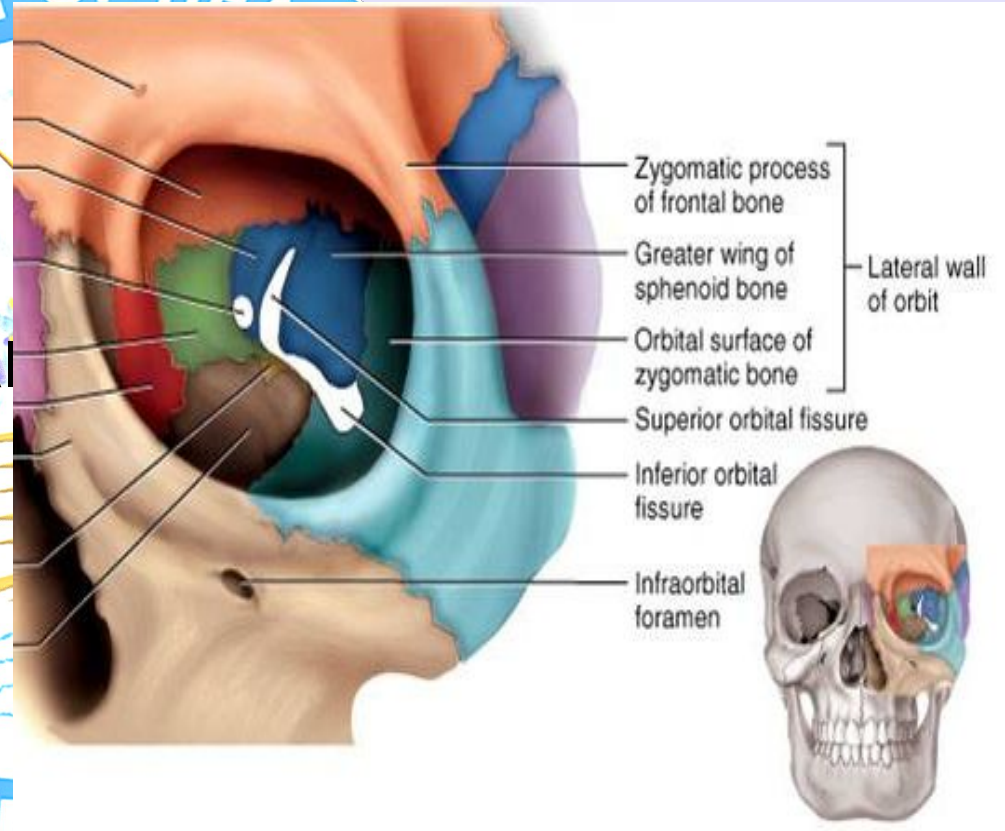
Formed :

- Anteriorly by orbital surface of the frontal process of the zygomatic bone

(Separates orbit from temporal fossa)

- Posteriorly by ant. Surface of the greater wing of the sphenoid bone

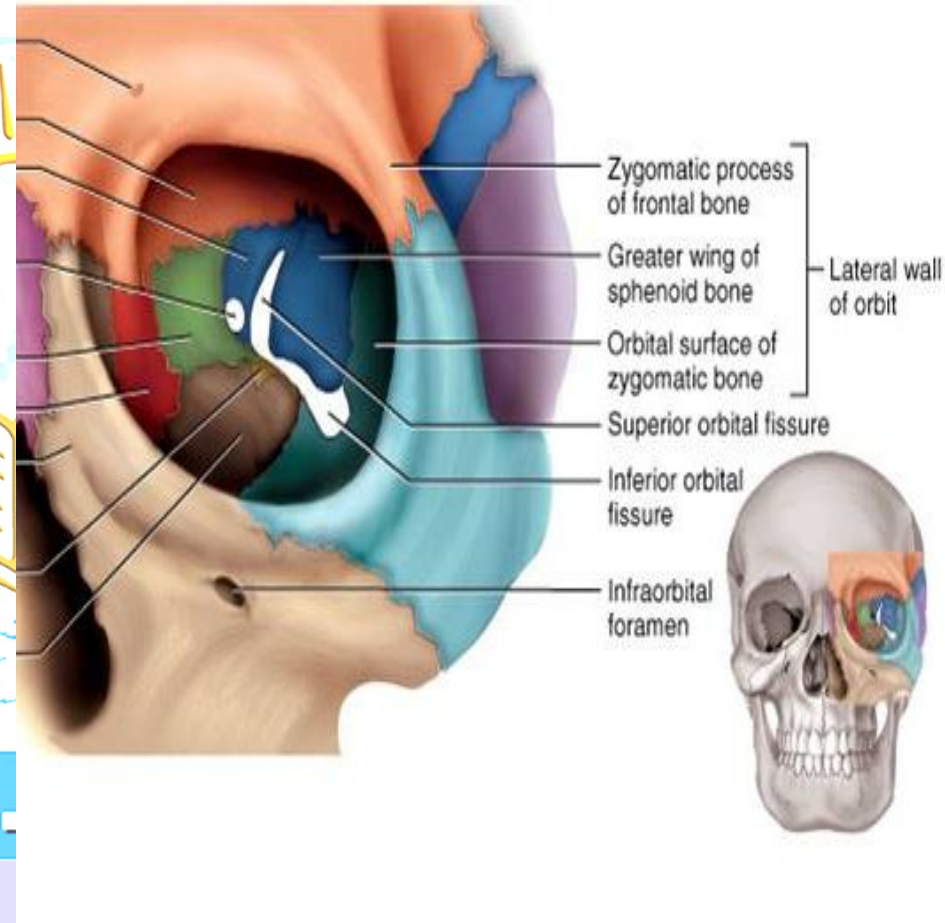
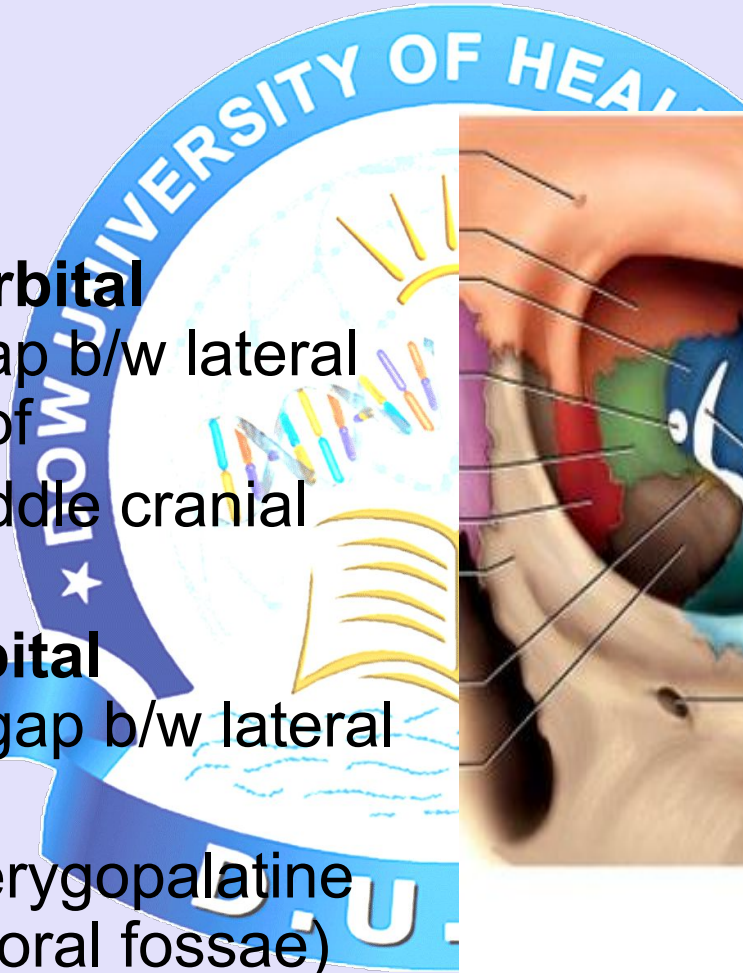
(Separates orbit from middle cranial fossa)



Contd...

FEATURES:

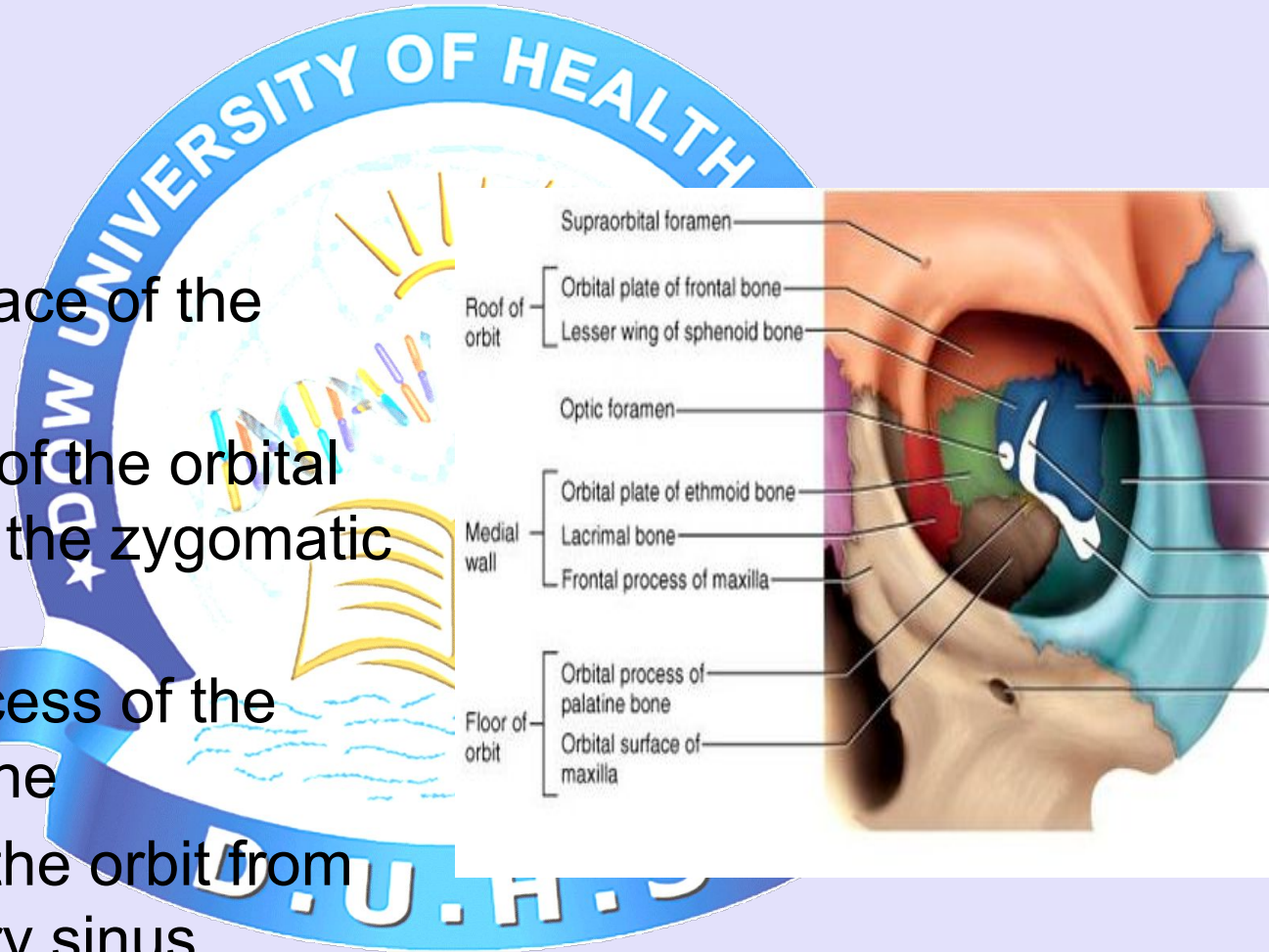
- **Superior orbital fissure**...gap b/w lateral wall and roof
(leads to middle cranial fossa)
- **Inferior orbital fissure**....gap b/w lateral wall & floor
(Leads to pterygopalatine & infratemporal fossae)



FLOOR

Formed by:

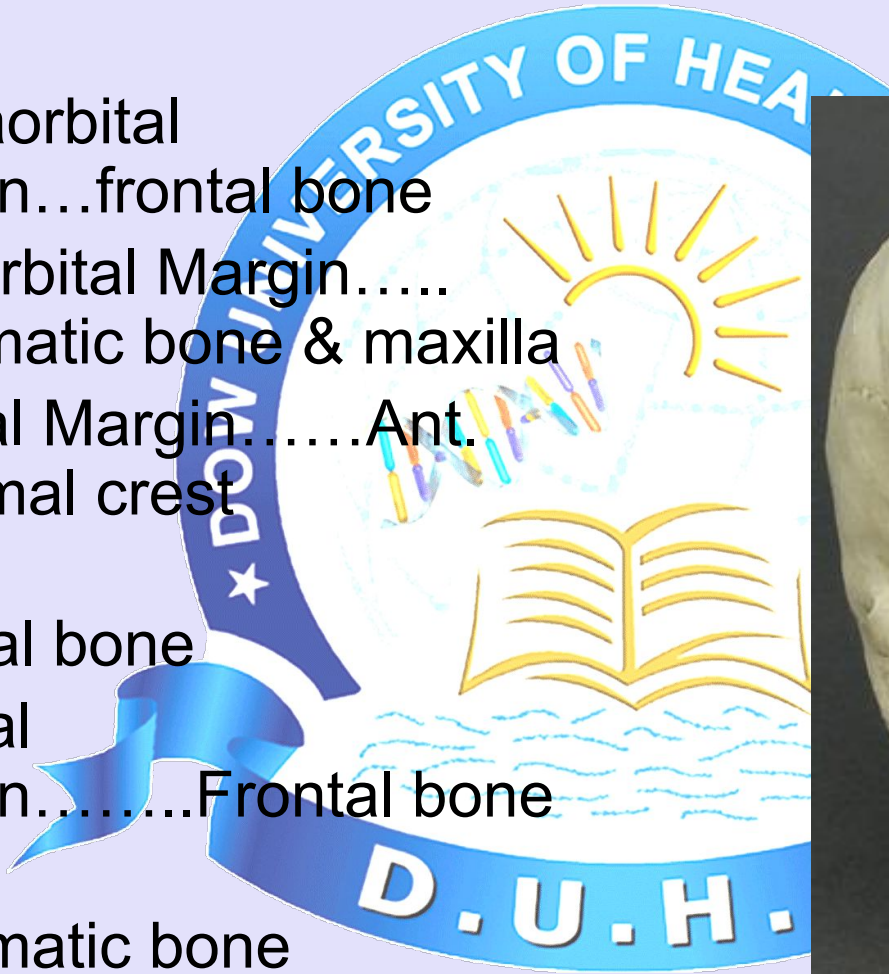
- Orbital surface of the maxilla
- Lower part of the orbital surfaces of the zygomatic bone
- Orbital process of the palatine bone
- Separates the orbit from the maxillary sinus



ORBITAL MARGINS

- Supraorbital Margin...frontal bone
- Infraorbital Margin.....zygomatic bone & maxilla
- Medial Margin.....Ant. Lacrimal crest

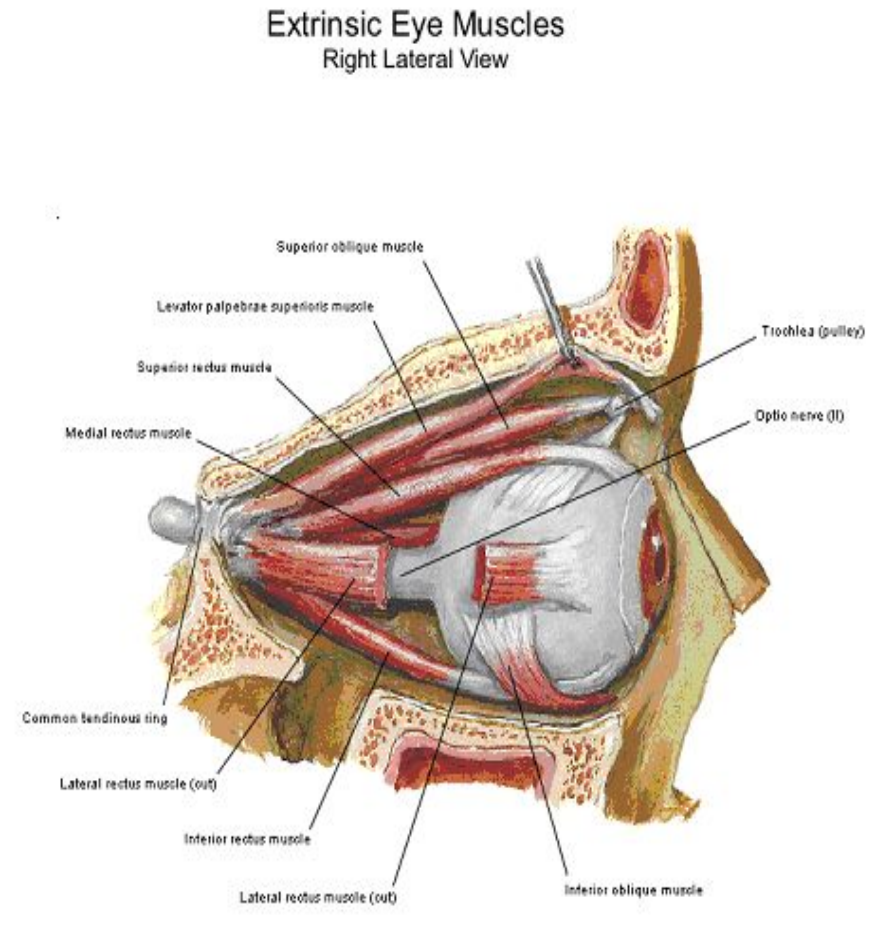
- Frontal bone
- Lateral Margin.....Frontal bone
- Zygomatic bone



Extraocular Muscles

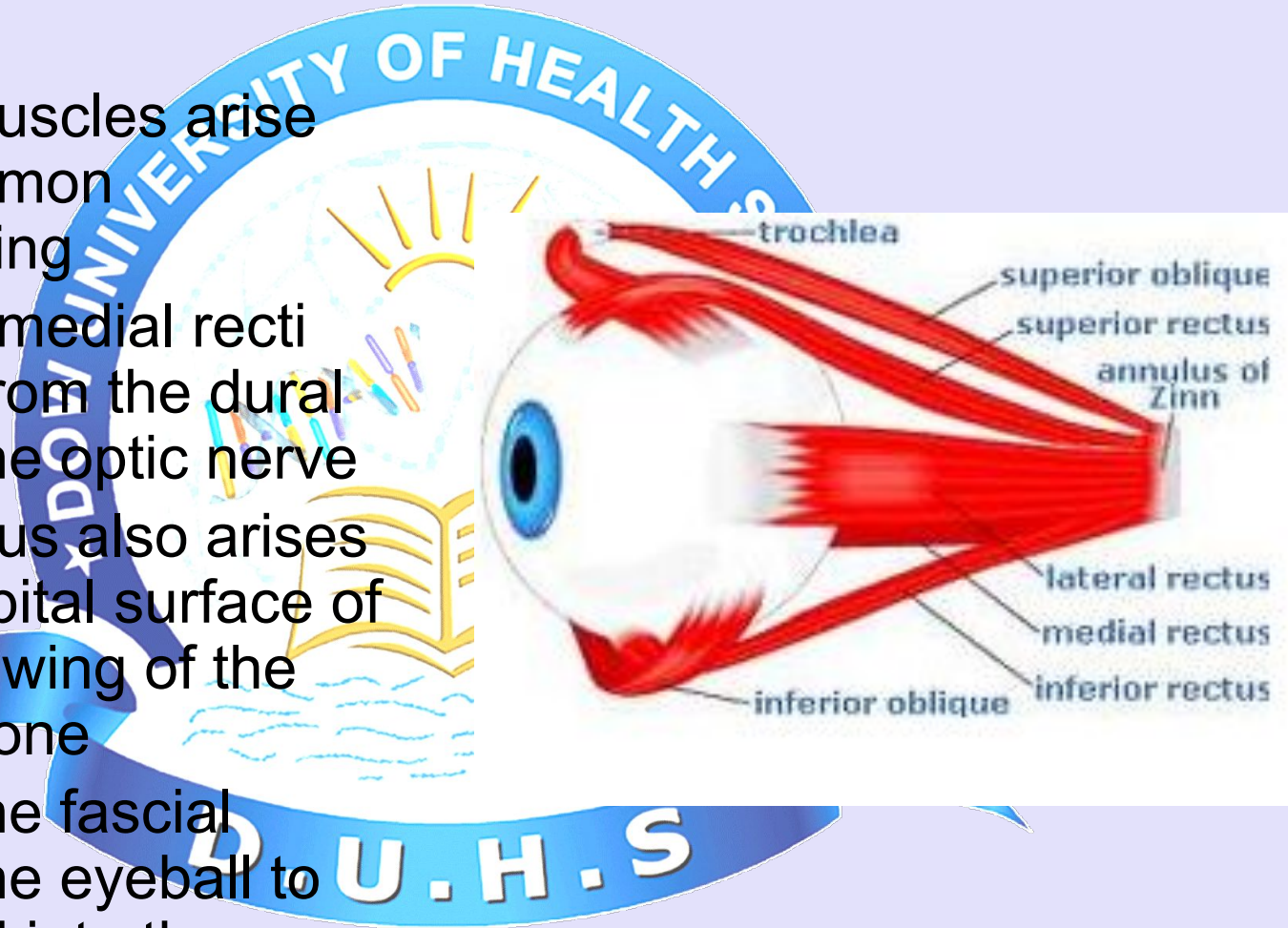
- Eyeball is moved by extrinsic or the extraocular muscles:

- ✓ Four recti
- ✓ Two obliqui
- ✓ Levator palpebrae superioris



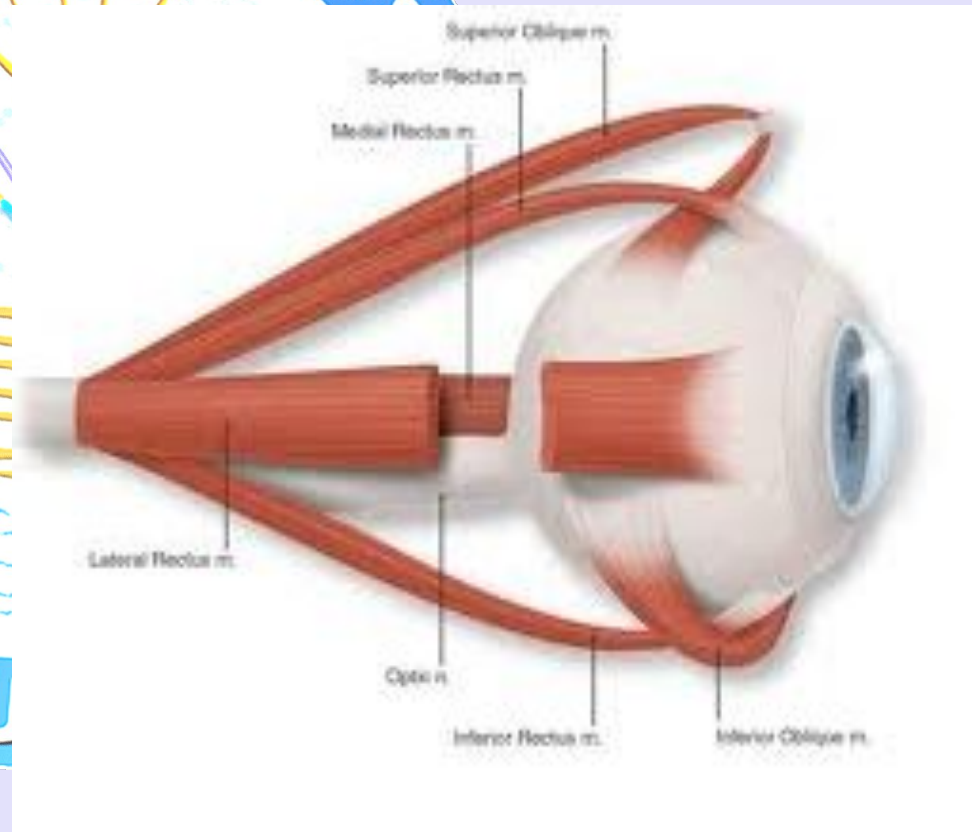
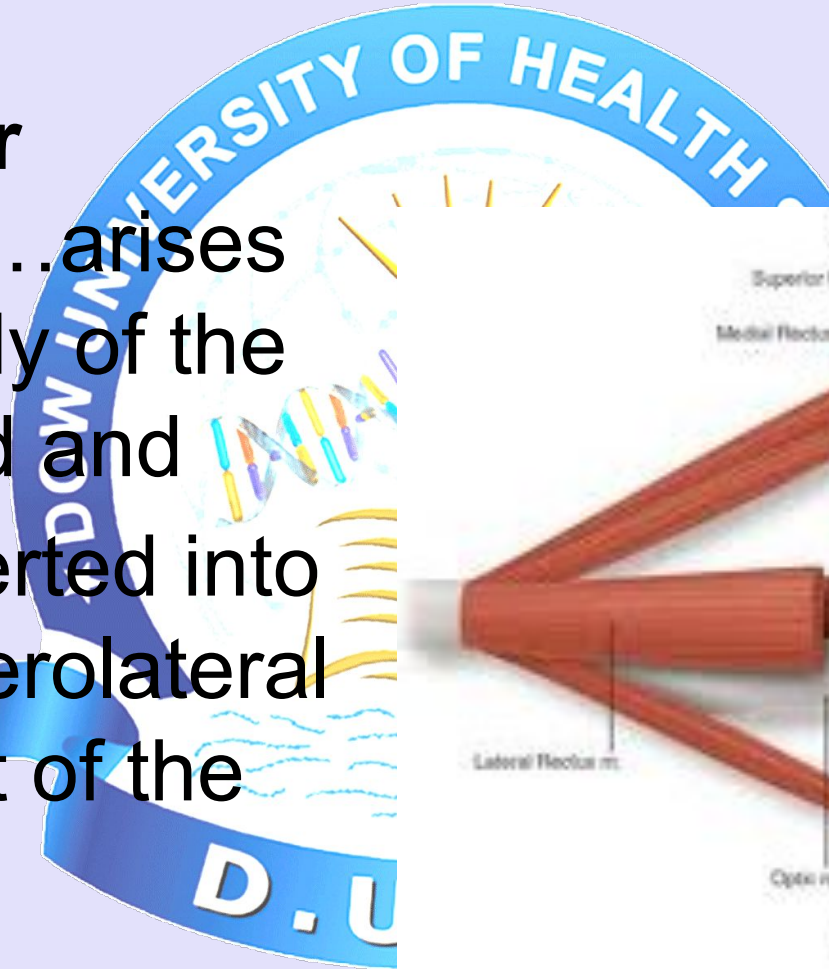
THE RECTI MUSCLES

- The recti muscles arise from a common tendinous ring
- ✓ Superior & medial recti also arise from the dural sheath of the optic nerve
- ✓ Lateral rectus also arises from the orbital surface of the greater wing of the sphenoid bone
- ✓ All pierce the fascial sheath of the eyeball to get inserted into the sclera



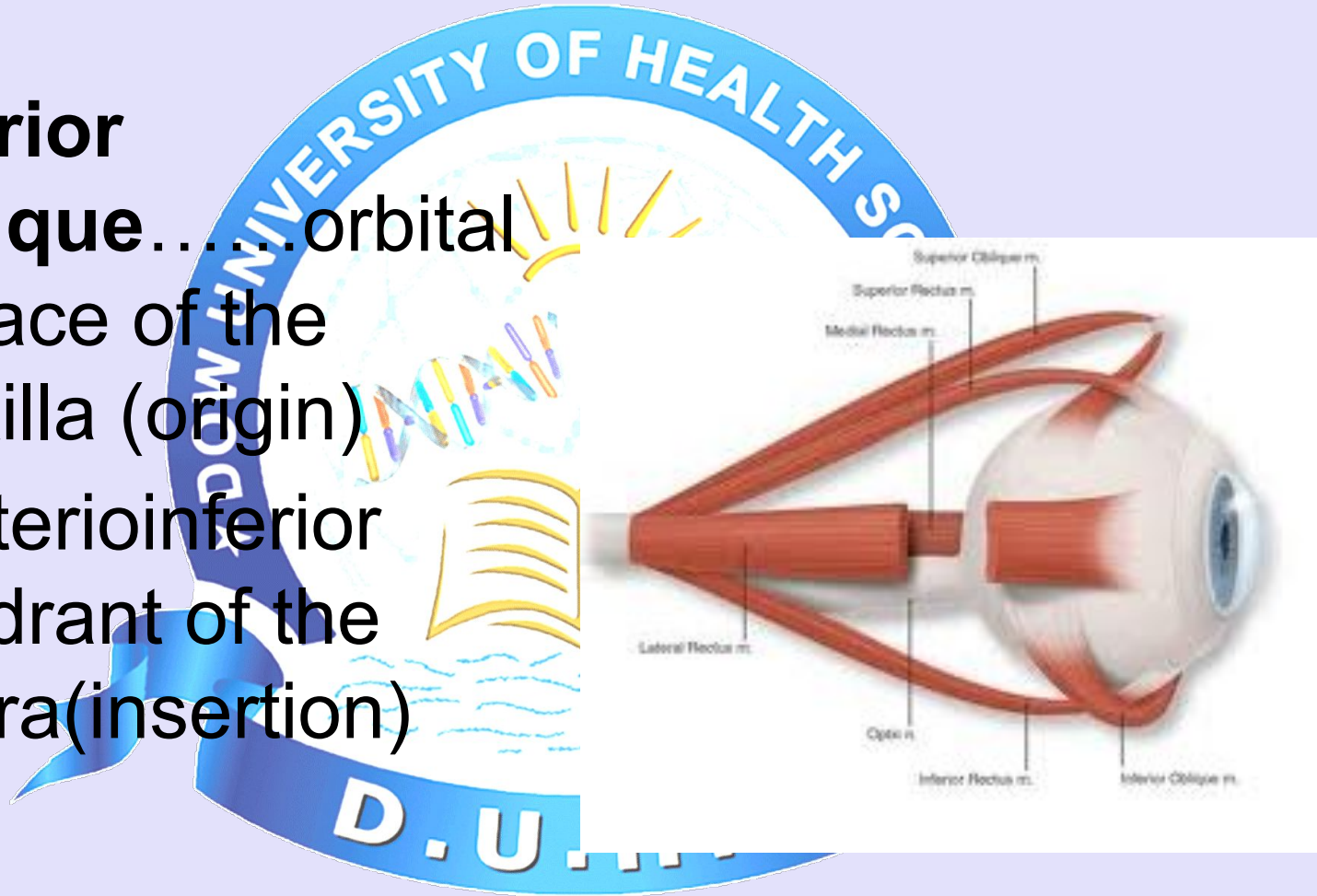
CONTD...

- **Superior oblique...** arises from body of the sphenoid and
- gets inserted into the posterolateral quadrant of the sclera



CONTD...

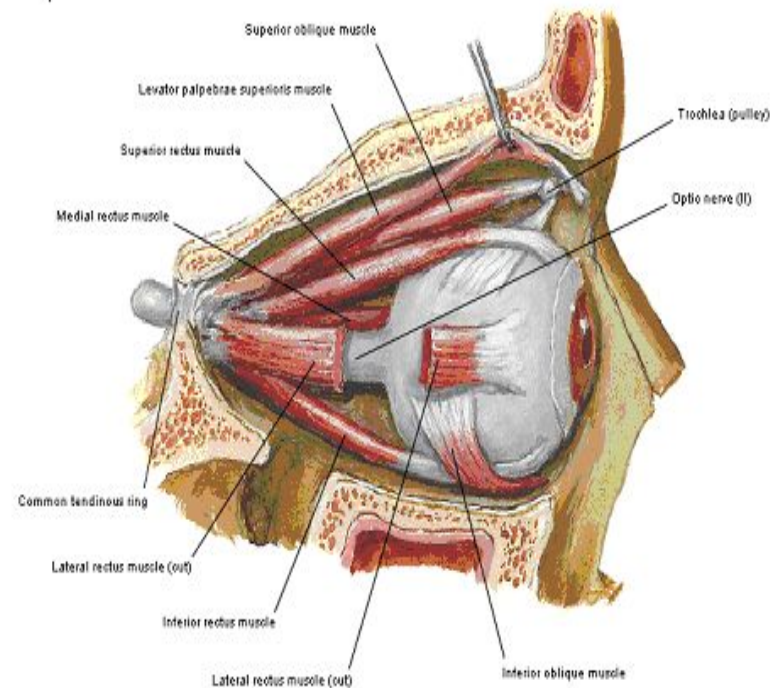
- **Inferior Oblique.....orbital surface of the maxilla (origin)**
- **Posterioinferior quadrant of the sclera(insertion)**

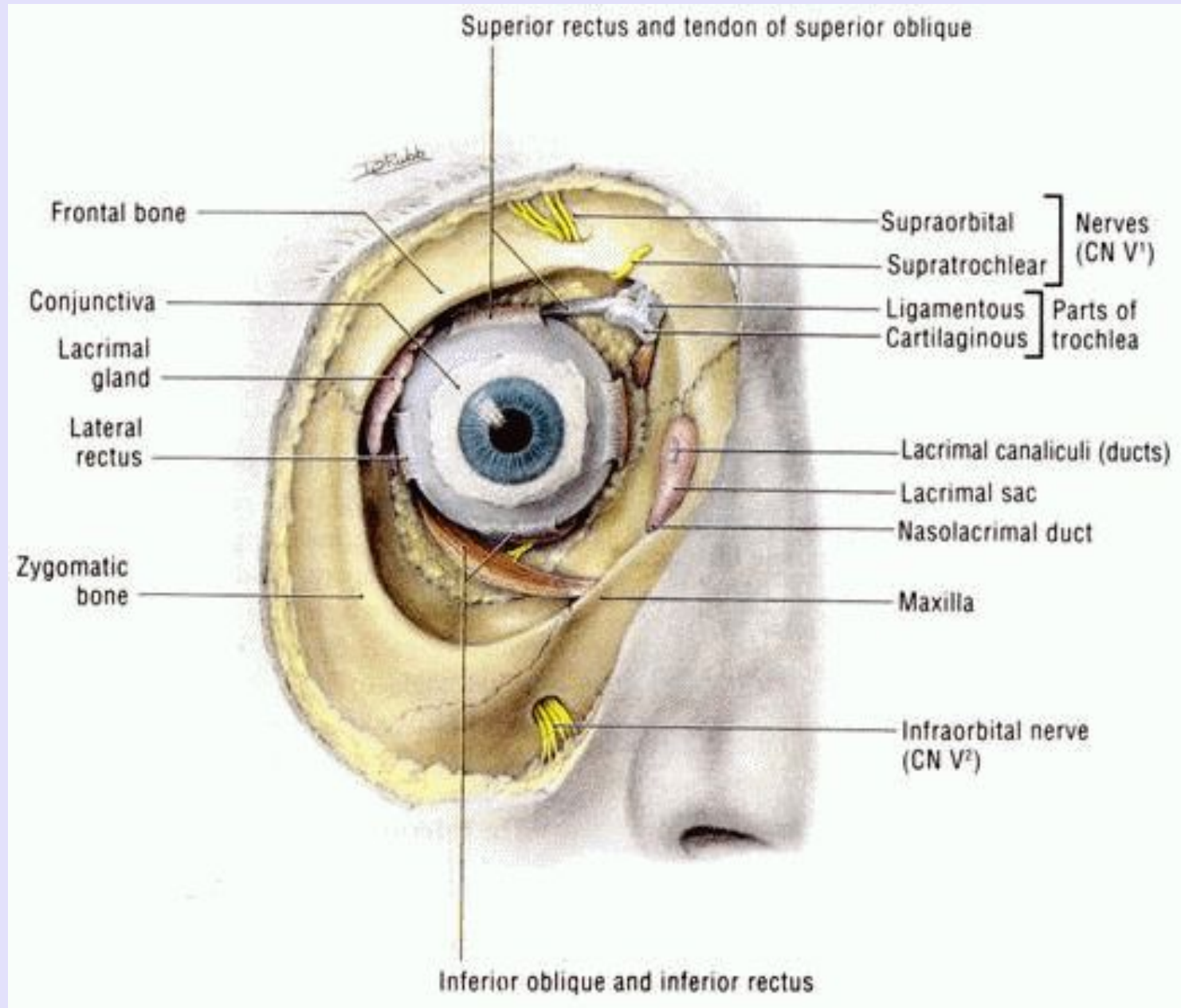


Contd....

- Levator palpebrae superioris...
- ✓ .arises from undersurface of the lesser wing of the sphenoid
- ✓ Splits into two lamellae
- ✓ Superior(voluntary) inserts into ant surface of superior tarsus and skin of upper eyelid
- ✓ Inferior(involuntary) inserts into upper margin of superior tarsus

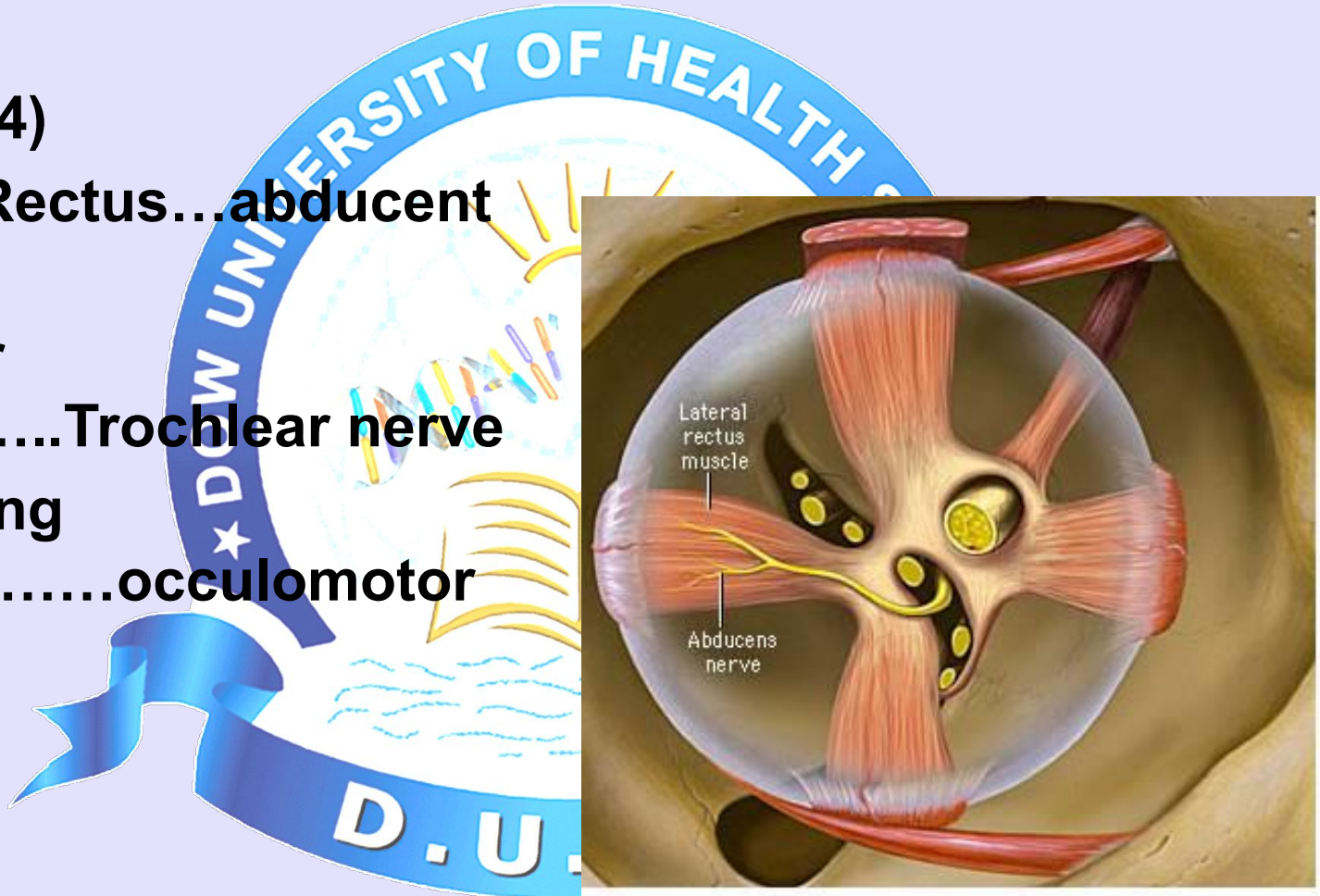
Extrinsic Eye Muscles
Right Lateral View





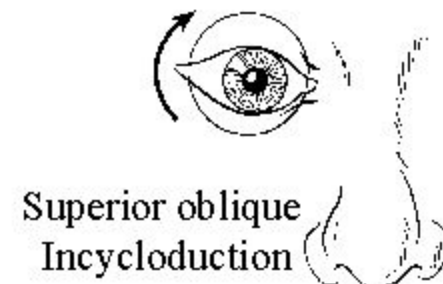
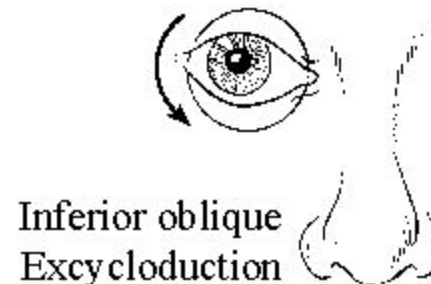
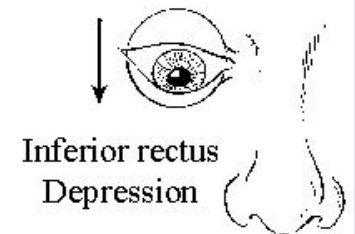
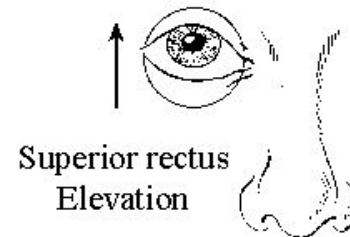
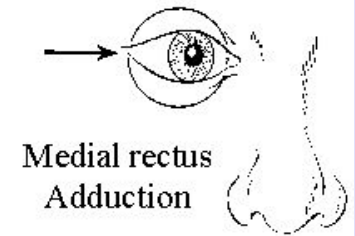
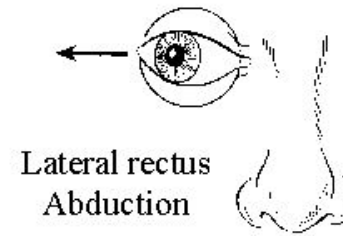
Nerve Supply

- (LR6 SO4)
- Lateral Rectus...abducent nerve
- Superior oblique.....Trochlear nerve
- Remaining muscles.....occulomotor nerve



ACTIONS

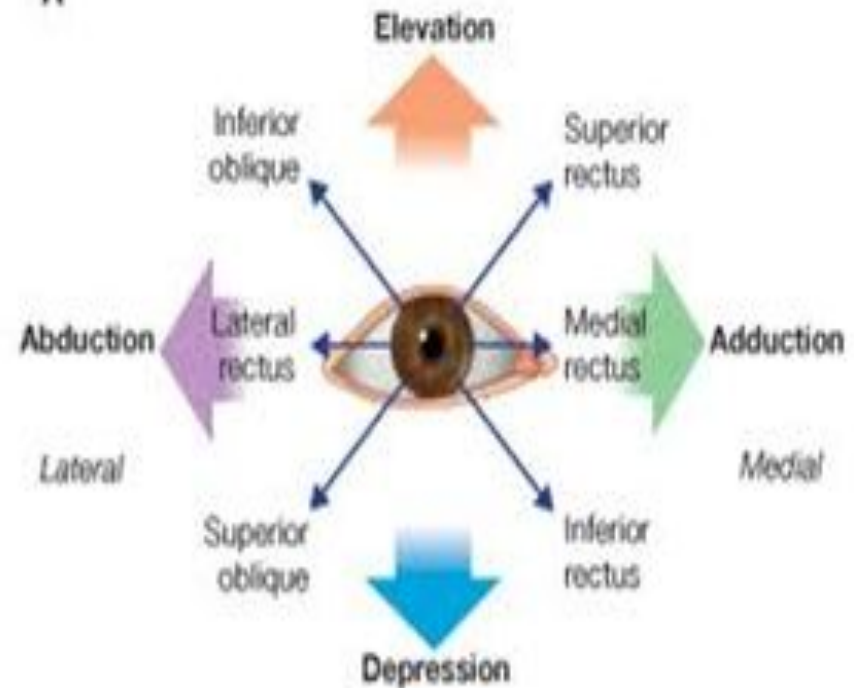
- Levator Palpabrae Superioris...Elevation of upper eyelid
- Superior rectus: Upward rotation
- Medial rotation
- Intortion
- Inferior rectus: Downward rotation
- Medial rotation
- Extortion



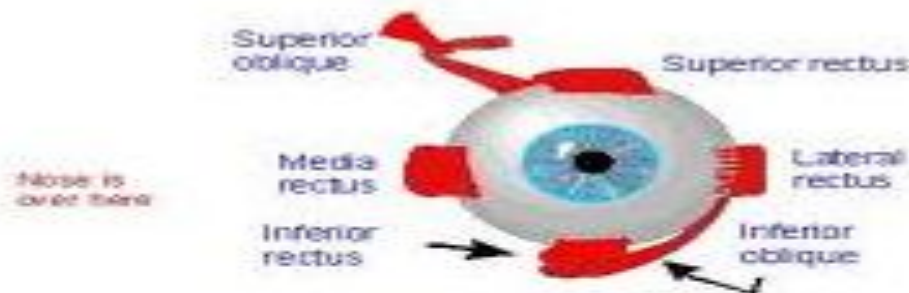
Contd...

- **Medial rectus:** Medial rotation
- **Lateral rectus :** Lateral rotation
- **Superior oblique:**
Downward rotation
Lateral rotation
- **Inferior oblique:** Upward
rotation

Lateral rotation
Extortion

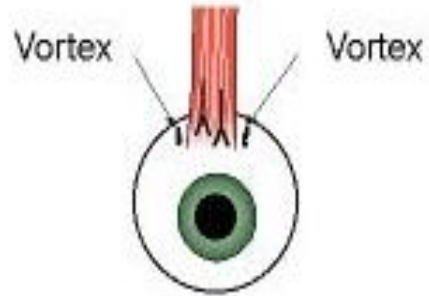


- Medial rectus moves the eye towards the nose
- Lateral rectus moves the eye away from the nose
- Superior rectus moves the eye up
- Inferior rectus moves the eye down
- Superior oblique rotates the eye so that the top of the eye moves towards the nose
- Inferior oblique rotates the eye so that the top of the eye moves away from the nose

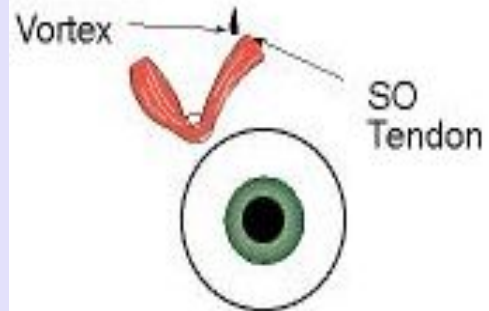


The Left Eye

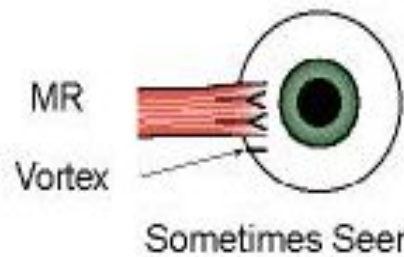
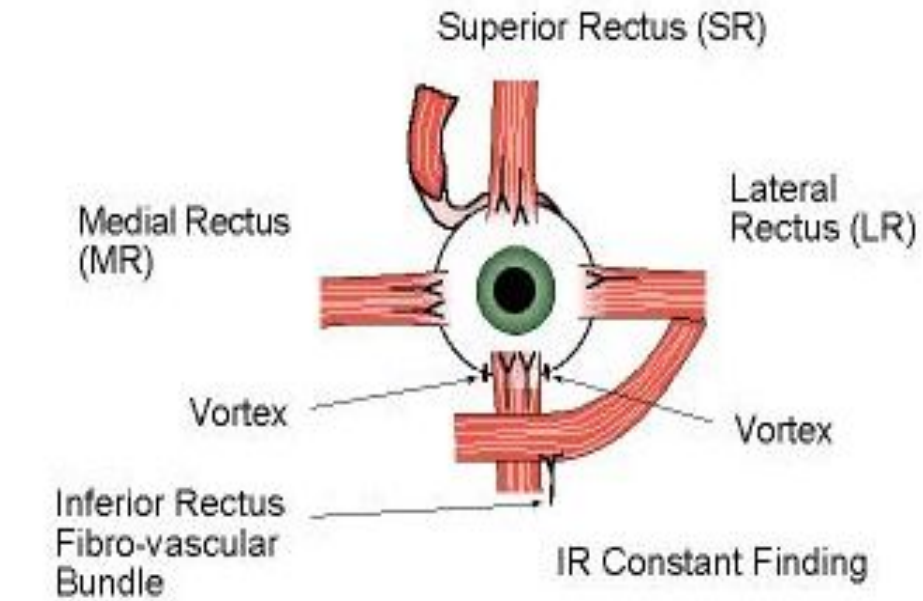
Inferior Rectus (IR)



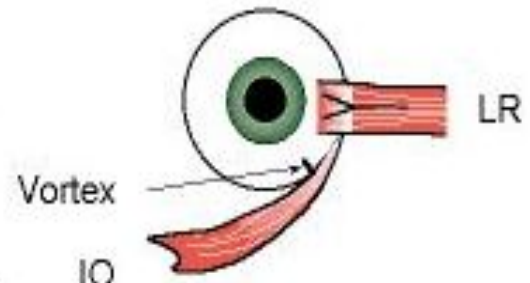
IR – Looking from above Surgeon's



Constant Finding



Sometimes Seen



Constant Finding

